

Myelodysplastic syndromes

Clonal stem-cell disorders with **ineffective hematopoiesis** → cytopenias, dysplasia, and variable AML risk · blasts <20% · everything collapses to **lower risk** (fix the cytopenias) vs **higher risk** (alter the natural history)

Compiled from the Encyclopedia MDS review note and Fellowship Anki deck · WHO 2022 / ICC · IPSS-M (Bernard, NEJM Evid 2022) · Bernard/Papaemmanuil molecular taxonomy (Blood 2024)

DIAGNOSIS

- **Triad:** (1) cytopenia + (2) morphologic **dysplasia in >10%** of a lineage + (3) **clonality** (cytogenetic or molecular). >90% show clonality on karyotype/NGS
- Normal marrow cellularity ≈ **100 – age**. Most MDS is **hypercellular** with ineffective hematopoiesis; a hypoplastic variant overlaps aplastic anemia
- **Dysplasia is not MDS without clonality**. Exclude mimics first: **B12, folate, copper** deficiency, HIV, alcohol, medications, post-chemotherapy marrow recovery
- † Hypoplastic MDS overlaps **aplastic anemia and PNH** — send flow for a PNH clone
- † Consider **VEXAS** (somatic **UBA1**) in older men with inflammatory features, macrocytic anemia and **vacuoles in myeloid and erythroid precursors**

CLASSIFICATION (WHO 2022 / ICC)

Genetically defined	SF3B1 (favorable, ring sideroblasts) · isolated del(5q) (lenalidomide-responsive) · biallelic TP53 (poor)
Blast categories	MDS-IB1 = 5–9% blasts · MDS-IB2 = 10–19%
The AML border	At 10–19% blasts with a defining lesion, ICC calls it MDS/AML (managed as AML) while WHO calls it MDS-IB2
† del(5q) phenotype	Macrocytic anemia, normal or high platelets , hypolobated megakaryocytes, female predominance

RISK STRATIFICATION

IPSS-R (2012)	Cytogenetic risk, marrow blast %, hemoglobin, platelets, ANC → 5 groups
IPSS-M (2022)	Keeps blasts, Hgb, platelets, cytogenetics · drops ANC · adds 31 gene mutations → 6 groups
Mutation weights	TP53-multihit highest risk · SF3B1 favorable · adverse: ASXL1, RUNX1, SRSF2, U2AF1, EZH2, FLT3, KMT2A-PTD

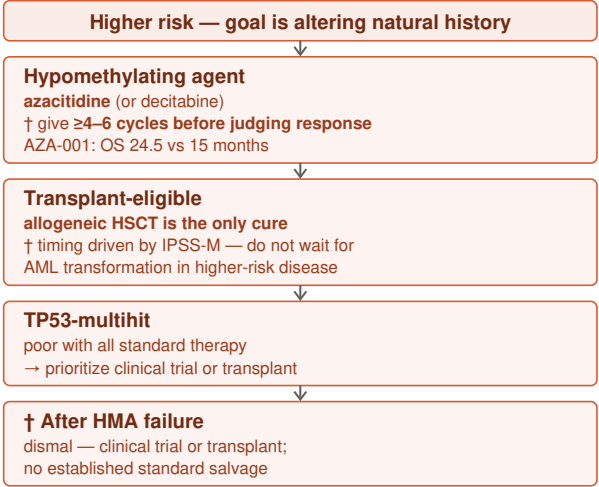
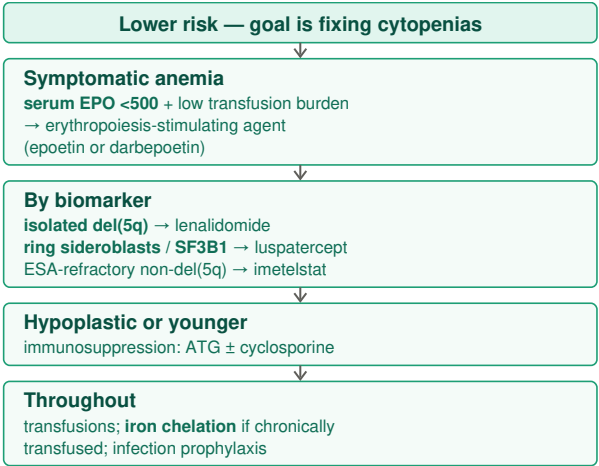
Calculator at mds-risk-model.com. IPSS-M still outperforms a purely molecular-subgroup model, so use it for risk scoring.

Molecular taxonomy (Bernard/Papaemmanuil, Blood 2024).
3,233 patients, 152-gene panel → **16 molecular groups** covering 86%, median OS ranging **0.9–8.2 years**.

Favorable: SF3B1 (14%, most common, OS ~6 yr) · del(5q) · CCUS-like · ZRSR2 · driver-negative. **Monocytic:** bi-TET2 (13%, CMML overlap). **Aggressive:** AML-like (OS ~0.9 yr, 40% 2-yr AML transformation) · TP53-complex · IDH-STAG2 · –7/SETBP1 · EZH2-ASXL1 · BCOR. **DDX41:** high blasts and high transformation **but no excess mortality**.

Take-home: the prognostic weight of **blast %** depends on genetic subtype — but **IPSS-M still outperforms** a groups-only model, so the taxonomy informs biology rather than replacing risk scoring.

TREATMENT PATHWAY



BIOMARKER / SETTING → THERAPY

Setting	Therapy	Key trial
Lower risk, EPO <500, low transfusion burden	Erythropoiesis-stimulating agent	—
Lower risk, isolated del(5q)	Lenalidomide	MDS-003 / MDS-004
Lower risk, transfusion-dependent, ring sideroblasts / SF3B1	Luspatercept — activin receptor II ligand trap, improves late-stage erythropoiesis	MEDALIST; COMMANDS (beat epoetin alfa first-line)
Lower risk, ESA-refractory, non-del(5q)	Imetelstat — telomerase inhibitor	IMerge (FDA-approved 2024)
Hypoplastic / younger lower risk	Immunosuppression (ATG ± cyclosporine)	—
Higher risk	Azacitidine or decitabine	AZA-001 — OS benefit
Higher risk, transplant-eligible	Allogeneic HSCT — the only curative option	—
TP53-multihit	Poor with everything → trial or transplant	—

TRIALS

CALGB 9221 (2002)	The first RCT of azacitidine vs supportive care — higher response and delayed transformation to AML or death . The disease-modifying signal (Silverman, JCO)
AZA-001 (2009)	Practice-defining — overall survival benefit in higher-risk MDS, median OS 24.5 vs 15 months (Fenaux, Lancet Oncol)
† Decitabine (D-0007, EORTC 06011)	Improved response and delayed AML progression but no OS benefit . Caveat: decitabine was capped at 6 cycles (median 4) while azacitidine continued to progression (median 9). Practically, azacitidine is the HMA with an OS benefit
MDS-003 / MDS-004	Lenalidomide in transfusion-dependent lower-risk del(5q) — ~67% transfusion independence (List, NEJM 2006)
MEDALIST (2020)	Luspatercept after ESA failure, ring sideroblasts / SF3B1 — transfusion independence 38% vs 13%
COMMANDS (2023)	Luspatercept beat epoetin alfa first-line — ~59% vs ~31%
IMerge (2024)	Imetelstat in ESA-refractory non-del(5q) — 8-week transfusion independence 40% vs 15%
† BMT CTN 1102	Donor vs no-donor in higher-risk MDS age 50–75 — 3-yr OS 47.9% vs 26.6% . The evidence for transplanting older patients (Nakamura, JCO 2021)
† VERONA (2025)	Venetoclax + azacitidine in higher-risk MDS missed its OS endpoint (22.2 vs 21.7 mo, HR 0.91) despite higher response rates — ven/aza is standard in AML but not MDS